CS342 · VISUAL PROGRAMMING

Midterm 461 Solutions

Time Converter

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01 What is required

Step	Marks	In code
1	5	**public class TimeConverter extends JFrame**
2	3	4 JTextField, 4 JLabel and 2 JButton as attributes
3	6	try, throw and 3 catch blocks, each with showMessageDialog
4	6	Inner classes BConvert and BCancel, each implements ActionListener
5	—	**setSize(450, 100)**

The reverse formula	With 12,500
ho = s / 3600	12500 ÷ 3600 = 3
min = (s % 3600) / 60	1700 ÷ 60 = 28
sec = s % 60	12500 mod 60 = 20

02 The full solution

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
public class TimeConverter extends JFrame {
    private JTextField tSec, tHo, tMin, tS;
    private JLabel l1, l2, l3, l4;
    private JButton bConvert, bCancel;
    public TimeConverter() {
        setTitle("Time Converter");
        setSize(450, 100);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        JPanel p = (JPanel) getContentPane();
        p.setLayout(new FlowLayout());
        tSec = new JTextField(10);
        l1   = new JLabel("seconds  =");
        tHo  = new JTextField(3);
        l2   = new JLabel("h :");
        tMin = new JTextField(3);
        l3   = new JLabel("m :");
        tS   = new JTextField(3);
        l4   = new JLabel("s");
        bConvert = new JButton("Convert");
        bCancel  = new JButton("Cancel");
        p.add(tSec); p.add(l1);
        p.add(tHo);  p.add(l2);
        p.add(tMin); p.add(l3);
        p.add(tS);   p.add(l4);
        p.add(bConvert); p.add(bCancel);
        bConvert.addActionListener(new BConvert());
        bCancel.addActionListener(new BCancel());
    }
```

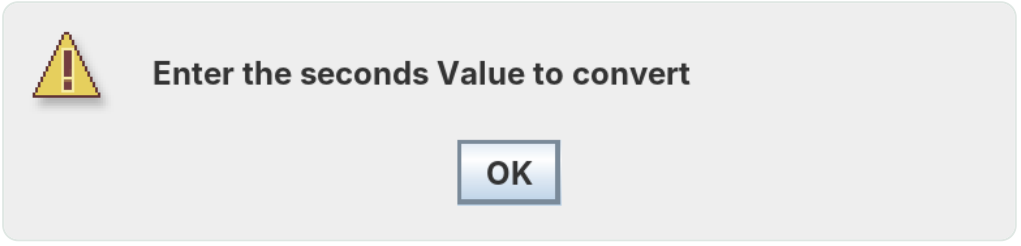
```
    class BConvert implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            try {
                String txt = tSec.getText();
                if (txt.equals("")) throw new EmptyFieldException();
                int s = Integer.parseInt(txt);
                if (s < 0 || s > 86400) throw new InvalidSecondsException();
                int ho  = s / 3600;
                int min = (s % 3600) / 60;
                int sec = s % 60;
                tHo.setText("" + ho);
                tMin.setText("" + min);
                tS.setText("" + sec);
            }
            catch (EmptyFieldException ex) {
                JOptionPane.showMessageDialog(null, "Enter the seconds Value to
convert",
                        "Empty Input", JOptionPane.WARNING_MESSAGE);
            }
            catch (NumberFormatException ex) {
                JOptionPane.showMessageDialog(null, "Enter the seconds as a numerical
Value",
                        "Error: Input Format", JOptionPane.ERROR_MESSAGE);
            }
            catch (InvalidSecondsException ex) {
                JOptionPane.showMessageDialog(null, "Enter the seconds Value in the
range [0..86400].",
                        "Error: Invalid Input Value", JOptionPane.ERROR_MESSAGE);
            }
        }
    }
    class BCancel implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            tSec.setText("");
            tHo.setText("");
```

```
            tMin.setText("");
            tS.setText("");
        }
    }
    public static void main(String[] args) {
        TimeConverter f = new TimeConverter();
        f.setVisible(true);
    }
}
```

The exception classes are already defined per the question. Written out: class EmptyFieldException extends Exception and the same for InvalidSecondsException, each with an empty body.

03 What the user sees

Input	Result
12500	3 · 28 · 20
(empty)	Empty Input warning
abc	Error: Input Format
90000 or −5	Error: Invalid Input Value





Enter the seconds as a numerical Value

OK



Enter the seconds Value in the range [0..86400].

OK